

Revision Change Record

Comparison baseline: upstream commit 4d636d0, 17 September 2026. This is a local-edit record, not the highlighted comparison against the originally reviewed submission.

diff --git a/paper/ieee-paper/paper-access.tex b/paper/ieee-paper/paper-access.tex

--- a/paper/ieee-paper/paper-access.tex

+++ b/paper/ieee-paper/paper-access.tex

@@ -13,6 +13,27 @@

AFTER: \hypersetup{hidelinks}

AFTER: \pdfminorversion=7

AFTER: \makeatletter

AFTER: % The template overlays full-width footer boxes; use explicit zero-width overlaps.

AFTER: \newcommand{\fixaccessfooter}[1]{%

AFTER: \patchcmd{#1}{\hbox to 0pc}{\rlap{\fixaccessfooter{#1}}}%

AFTER: }

AFTER: \fixaccessfooter{\ps@titlepage}

AFTER: \fixaccessfooter{\ps@headings}

AFTER: \ps@headings

AFTER: \makeatother

AFTER: % Separate the abstract width from the template's aliased title-width register.

AFTER: \newlength{\accessabstractwidth}

AFTER: \AtBeginDocument{%

AFTER: \setlength{\accessabstractwidth}{\dimexpr\titlewidth-0.5em-1.5mm\relax}%

AFTER: \let\abstractwidth\accessabstractwidth

AFTER: \input{t1formata.fd}%

AFTER: \DeclareFontShape{T1}{formata}{m}{sl}{t1-formata-mediumitalic}{}%

AFTER: \makeatletter\input{t1pcr.fd}\makeatother

AFTER: \DeclareFontShape{T1}{pcr}{n}{n}{pcrr8t}{}%

AFTER: }

@@ -127,7 +148,15 @@ Given a set of wire segments $\mathcal{W} = \{w_1, \ldots, w_m\}$ with endpoints

BEFORE: The tolerances are manually configured scale-relative rules with fixed pixel clamps:

$\tau_{\text{pin}} = \min(60, \max(24, 0.62s))$, $\tau_{\text{join}} = \min(28, \max(11, 0.30s))$, and $\tau_t = \min(20, \max(8, 0.20s))$, in pixels. Here s is the characteristic component-bounding-box diagonal scale; the implementation uses the middle sorted diagonal (the upper middle for an even count), falling back to a wire-length scale when component geometry is unavailable or degenerate. The multipliers and clamps are fixed, not learned. Because s is a single scalar per image, intra-image size variance is

not modeled: a schematic mixing a large IC with small discrete symbols shares one tolerance per rule. Extreme-size performance was not evaluated by controlled image rescaling.

AFTER: The tolerances are manually configured scale-relative rules with fixed pixel clamps (in pixels):

AFTER: $\begin{equation*}$

AFTER: $\begin{aligned}$

AFTER: $\tau_{\text{pin}} \&= \min(60, \max(24, 0.62s)), \backslash\backslash$

AFTER: $\tau_{\text{join}} \&= \min(28, \max(11, 0.30s)), \backslash\backslash$

AFTER: $\tau_t \&= \min(20, \max(8, 0.20s)).$

AFTER: $\end{aligned}$

AFTER: $\end{equation*}$

AFTER: Here s is the characteristic component-bounding-box diagonal scale; the implementation uses the middle sorted diagonal (the upper middle for an even count), falling back to a wire-length scale when component geometry is unavailable or degenerate. The multipliers and clamps are fixed, not learned. Because s is a single scalar per image, intra-image size variance is not modeled: a schematic mixing a large IC with small discrete symbols shares one tolerance per rule. Extreme-size performance was not evaluated by controlled image rescaling.

@@ -241,7 +270,7 @@ The synthetic results above measure robustness under an authored error model. Fo

BEFORE: $\begin{table}[htbp]$

AFTER: $\begin{table*}[t]$

@@ -259,7 +288,7 @@ Directional preference disabled & 0.816 & 352/27/132 & 0.890 & 0.919 & 0.864 \\\

BEFORE: $\end{table}$

AFTER: $\end{table*}$

@@ -309,7 +338,7 @@ With annotated component boxes held fixed, replacing detected wires by the datas

BEFORE: $\includegraphics[width=0.82\textwidth]{figures/complexity_histogram.pdf}$

AFTER: $\includegraphics[width=\columnwidth]{figures/complexity_histogram.pdf}$

@@ -425,8 +454,8 @@ on the CGHD-1152 dataset~\cite{bayer2021cghd}.

BEFORE: $\text{\textit{gao2026memristor}}$ S.~Gao, F.~Shi, X.~Xu, J.~Chen, H.~H.-C.~Lu, et al., A memristor-based neural network circuit with classical conditioning and fear generalization," $\text{\textit{IEEE Trans. Consumer Electron.}}$, 2026, doi: 10.1109/TCE.2026.3663359.

BEFORE: $\text{\textit{gao2026memristive}}$ S.~Gao, Y.~Song, Y.~Cao, H.~H.-C.~Lu, L.~Qin, Y.~Zhang, and J.~Mou, Biologically plausible memristive decision-making circuit for adaptive control in industrial autonomous navigation," $\text{\textit{IEEE Trans. Ind. Informat.}}$, vol.~22, pp.~4801--4812, 2026, doi: 10.1109/TII.2026.3668794.

AFTER: $\text{\textit{gao2026memristor}}$ S.~Gao et al., A memristor-based neural network circuit with classical conditioning and fear generalization," $\text{\textit{IEEE Trans. Consumer Electron.}}$, vol.~72, no.~2, pp.~3213--3224, May 2026, doi: 10.1109/TCE.2026.3663359.

AFTER: $\text{\textit{gao2026memristive}}$ S.~Gao et al., Biologically plausible memristive decision-making circuit for adaptive control in industrial autonomous navigation," $\text{\textit{IEEE Trans. Ind. Informat.}}$, vol.~22, no.~6, pp.~4801--4812, Jun. 2026, doi:

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--- a/paper/ieee-paper/paper-build.tex

+++ b/paper/ieee-paper/paper-build.tex

@@ -18,6 +18,7 @@

AFTER: \hypersetup{hidelinks}

@@ -103,7 +104,15 @@ Given a set of wire segments $\mathcal{W} = \{w_1, \dots, w_m\}$ with endpoints

BEFORE: The tolerances are manually configured scale-relative rules with fixed pixel clamps:

$\tau_{\text{pin}} = \min(60, \max(24, 0.62s))$, $\tau_{\text{join}} = \min(28, \max(11, 0.30s))$, and $\tau_t = \min(20, \max(8, 0.20s))$, in pixels. Here s is the characteristic component-bounding-box diagonal scale; the implementation uses the middle sorted diagonal (the upper middle for an even count), falling back to a wire-length scale when component geometry is unavailable or degenerate. The multipliers and clamps are fixed, not learned. Because s is a single scalar per image, intra-image size variance is not modeled: a schematic mixing a large IC with small discrete symbols shares one tolerance per rule. Extreme-size performance was not evaluated by controlled image rescaling.

AFTER: The tolerances are manually configured scale-relative rules with fixed pixel clamps (in pixels):

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AFTER: $\tau_{\text{join}} \&= \min(28, \max(11, 0.30s)), \backslash$

AFTER: $\tau_t \&= \min(20, \max(8, 0.20s)).$

AFTER: \end{aligned}

AFTER: \end{equation*}

AFTER: Here s is the characteristic component-bounding-box diagonal scale; the implementation uses the middle sorted diagonal (the upper middle for an even count), falling back to a wire-length scale when component geometry is unavailable or degenerate. The multipliers and clamps are fixed, not learned. Because s is a single scalar per image, intra-image size variance is not modeled: a schematic mixing a large IC with small discrete symbols shares one tolerance per rule. Extreme-size performance was not evaluated by controlled image rescaling.

@@ -217,7 +226,7 @@ The synthetic results above measure robustness under an authored error model. Fo

BEFORE: \begin{table}[htbp]

AFTER: \begin{table*}[t]

@@ -235,7 +244,7 @@ Directional preference disabled & 0.816 & 352/27/132 & 0.890 & 0.919 & 0.864 \backslash

BEFORE: \end{table}

AFTER: \end{table*}

@@ -285,7 +294,7 @@ With annotated component boxes held fixed, replacing detected wires by the datas

BEFORE: \includegraphics[width=0.82\textwidth]{figures/complexity_histogram.pdf}

AFTER: \includegraphics[width=\columnwidth]{figures/complexity_histogram.pdf}

@@ -402,8 +411,8 @@ on the CGHD-1152 dataset~\cite{bayer2021cghd}.

BEFORE: \bibitem{gao2026memristor} S.~Gao, F.~Shi, X.~Xu, J.~Chen, H.~H.-C.~lu, et al., A memristor-based neural network circuit with classical conditioning and fear generalization," \textit{IEEE Trans. Consumer Electron.}, 2026, doi: 10.1109/TCE.2026.3663359.

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AFTER: \bibitem{gao2026memristor} S.~Gao et al., A memristor-based neural network circuit with classical conditioning and fear generalization," \textit{IEEE Trans. Consumer Electron.}, vol.~72, no.~2, pp.~3213--3224, May 2026, doi: 10.1109/TCE.2026.3663359.

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diff --git a/paper/ieee-paper/review_artifacts/RESPONSE_TO_REVIEWERS.md
b/paper/ieee-paper/review_artifacts/RESPONSE_TO_REVIEWERS.md

--- a/paper/ieee-paper/review_artifacts/RESPONSE_TO_REVIEWERS.md

+++ b/paper/ieee-paper/review_artifacts/RESPONSE_TO_REVIEWERS.md

@@ -8,7 +8,7 @@ Revised Article Title: "From Hand-Drawn Schematics to Structural Circuit Netli

BEFORE: Draft for author review. The concern/response/action structure follows the locally supplied IEEE Access response template; template conversion is complete. Final PDF page references, rendered response, figure reconciliation and submission-package approvals are pending; this source draft does not certify package readiness.

AFTER: Draft for author review. The concern/response/action structure follows the locally supplied IEEE Access response template. The local build adds page references from the accompanying Access PDF; see output/pdf/response_page_map.json and response-to-reviewers.pdf. The highlighted comparison uses commit 33f5e3d, selected by the author as the latest manuscript before August 13. Portal identity confirmation, author-field reconciliation and final package approvals remain pending.

@@ -68,7 +68,7 @@ Best regards,

BEFORE: **Author response:** We thank the reviewer for the suggestion and have cited both works in Related Work. We verified both records at abstract level (Gao et al., IEEE Trans. Consumer Electronics, Feb 2026; Gao et al., IEEE Trans. Industrial Informatics, vol. 22, 2026): both report neuromorphic memristive hardware for associative learning and navigation control, cited here as neighboring SPICE-deployment domains. Our comparison, benchmark claims, and narrowed scope are unchanged.

AFTER: **Author response:** We thank the reviewer for the suggestion and have cited both works in Related Work. We verified both records at abstract level and updated their publication details from the publisher-deposited DOI metadata (Gao et al., IEEE Trans. Consumer Electronics, vol. 72, no. 2, pp. 3213-3224, May 2026; Gao et al., IEEE Trans. Industrial Informatics, vol. 22, no. 6, pp. 4801-4812, June 2026): both report neuromorphic memristive hardware for associative learning and navigation control, cited here as neighboring SPICE-deployment domains. Our comparison, benchmark claims, and narrowed scope are unchanged.

@@ -112,7 +112,7 @@ Best regards,

BEFORE: **Author action:** Corrected tab:edge_ablation caption/row label and interpretation; removed activation/identical-connection explanations and the unsupported quantified occlusion benefit. Rewrote the method description to distinguish graph selection, mandatory slot/dummy assignment and guarded application. Table layout verification remains for T7.

AFTER: **Author action:** Corrected tab:edge_ablation caption/row label and interpretation; removed activation/identical-connection explanations and the unsupported quantified occlusion benefit. Rewrote the method description to distinguish graph selection, mandatory slot/dummy assignment and guarded application. The ablation table now spans both columns; the local rebuild removes its column overlap. Final coauthor page approval remains pending.